n = 83 D4 (K' = 4, 400 orbits) against M38: Tier 1 result report

Status (2026-11-08): result in. Verdict: improvement; every gate and control passes and the run is valid `[pcc:unverified]`. R = E2E(M38) / E2E(D4) = **2.0014 [1.8777, 2.1308]** over 60 alternating pairs, run 42425640847, Tier 1, instance scope. The design sections below are the frozen text, unchanged; they were frozen before any timing on the Tier 1 runner (commit `ebef3b260`). Every figure in them that comes from the shared cloud-agent VM is a local diagnostic, used only to size the design. The result section is added with the archived receipt.

Owner ruling, 2026-11-07 (recorded verbatim)

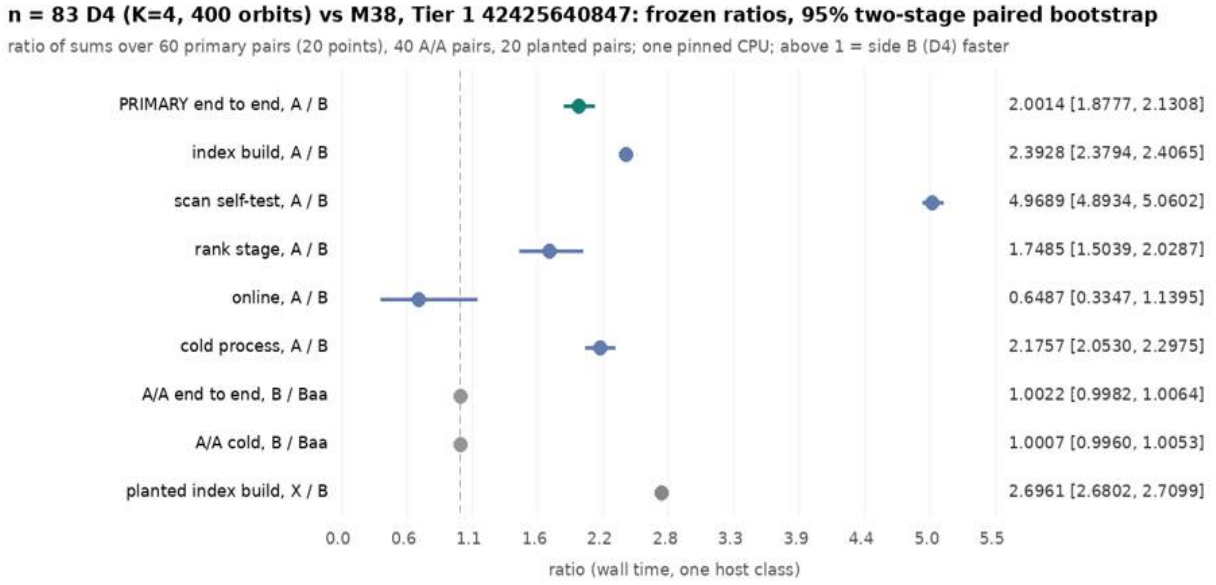
> Owner ruling, 2026-11-07, given in chat: a closure base with K' >= 4 > unknown-log columns counts as an index-calculus row. K' = 0 stays a > known-log reference row.

So D4 (K' = 4) is an index-calculus row, and this measurement is judged by the success criterion: a measured gain over the current n = 83 row. A K' = 0 base would be a known-log reference row (`koblitz-n83-knownlog-20261019`) and is not measured here. The ruling is also in `protocol.json` (`owner_ruling`) and in `docs/ic/BASELINES.md`, beside the known-log ruling-pending note.

Result

Run 42425640847 (workflow `n83-k4c400.yml`, push trigger, head `36866ad3c`, attempt 1, conclusion success) on one GitHub-hosted `ubuntu-24.04` runner (`GitHub Actions 1000197724`, image 20261101.360.1, **Intel Xeon Platinum 8370C**, execution CPU 0; the field-ops and field-ops-2 runs were on AMD EPYC 9V74). The schedule ran 50.7 minutes. The receipt is archived in `run_github_42425640847/` (`panel/runs.jsonl`, 2.4 MB, in the artifact store). The protocol and schedule are those of the frozen commit `ebef3b260`; commit `36866ad3c` changed only the workflow file (the artifact path and the store-pull step).

Verdict: improvement `[pcc:unverified]`. R = sum E2E(A) / sum E2E(B) = **2.0014 [1.8777, 2.1308]** over 60 alternating pairs (20 points x 3 pairs; two-stage paired bootstrap, 10,000 draws), A = M38 (K' = 16, 608 orbits), B = D4 (K' = 4, 400 orbits), the same binary `ic_online_fieldops_r770`. All 462 scheduled rows ran and are valid, none was skipped by the time guard, and every gate passes (`verify-stdout.json`): per base one index and layout digest set equal to its frozen one; Q0 equal to each base's frozen identity; identical relation digests, probe vectors, root calls and counters per D4 rank seed across kD4FO and kD4FOaa; rank tables equal to the frozen pre-checked tables; every IC row's pipeline names equal to its arm's; one scalar per point across both bases and rT, replayed in the oracle; placement, host identity and binaries unchanged. `verify_receipt.py` and `band_model.py` re-run on a copy of the receipt reproduce `panel/result.json`, `panel/result_table.md` and `band/band_model.json` byte for byte.



SVG).](figures/result.svg)

Primary, stages and controls (frozen rule)

stage (A = M38, B = D4)	side A mean (ms)	side B mean (ms)	A / B [95%]
index build (cold process)	5,116	2,138	2.3928 [2.3794, 2.4065]
scan self-test	1,093	220	4.9689 [4.8934, 5.0602]
rank stage (rank process)	3,815	2,182	1.7485 [1.5039, 2.0287]
online, one target	298	460	0.6487 [0.3347, 1.1395]
cold process	6,888	3,166	2.1757 [2.0530, 2.2975]
end to end (rank + cold)	10,703	5,348	**2.0014 [1.8777, 2.1308]**

The four stage rows are stage secondaries and never decide. The saving is in the setup (setup A / B 2.4353 [2.4198, 2.4516]: the index build over 2.4x fewer stored states, and the self-test 5.0x faster on the smaller Feistel domain) and in the rank stage (4 attempts instead of 16, each 2.4x more probes). The online stage is slower on D4 (A / B 0.6487, interval spanning 1: one target, one exponential sample per process). Process CPU time A / B 2.0715 [1.9855, 2.1626], process wall 2.0713; peak RSS 1,898 MB against 915 MB (A / B 2.0740) `[pcc:unverified]`.

- **A/A** (B against Baa, 40 pairs): end to end 1.0022 [0.9982, 1.0064] (interval contains 1), cold 1.0007 [0.9960, 1.0053] (inside [0.97, 1.03]), CPU time 1.0016 [0.9991, 1.0043].
- **Planted:** cD4FOX / cD4FO index build 2.6961 [2.6802, 2.7099] (lower bound above 1); the planted processes kept the frozen layout digests and checked 2 x stored states roots.
- **Probe law** (estimate, band [0.80, 1.20], both gated over at least 64 attempts): kD4FO **1.0999** over 240 attempts (bootstrap interval [0.9383, 1.2634]; its upper end is above the band, the gated estimate is inside), kM38FO 1.0018 [0.9233, 1.0897] over 960 attempts. The risk named before the run (the prototype's 1.154) did not materialise as a failure, but D4's point estimate is 10% above the law and its interval reaches 1.26: the true law ratio on D4 is not tight. kD4FO ns per probe 121.4.
- **Order and spread:** A first 2.0361, B first 1.9691; pair positions 2.0007, 2.0307, 1.9736; per-point ratios from 1.58 to 2.38 (20 of 20 above 1). Geometric mean of the per-pair ratios 2.0242 [1.9049, 2.1519].
- **Observed variance against the power check** (`observed_cv_per_side`, per pair): D4 rank stage 0.415 (modelled 0.50), online 0.79 (1.2), E2E 0.176 (0.23); M38 rank stage 0.222 (0.21), online 1.08 (1.2), E2E 0.090 (0.076). The model over-stated the D4 spread, so the design had at least the power `analysis/power.json` gives (0.998 at a true 1.20x).

The first run of the same frozen design (log only)

Run 42420993318 (head `ebef3b260`, the frozen commit) ran the same protocol and schedule and finished its whole schedule. Its job log printed the verifier's output: decision `improvement`, E2E(M38) / E2E(D4) = **1.964 [1.841, 2.091]**, every check true (including the probe law and the 64-attempt gate), A/A and planted controls passing. This is as reported from the job log by the coordinator `[pcc:unverified]`: its artifact step named the field-ops receipt directory, so **no receipt was uploaded and nothing from that run is archived or re-verified here**. It is reported beside the archived run, not combined with it. The two estimates (1.964 and 2.0014) have overlapping intervals; they are two draws of the same design (same points, same seeds, same schedule, two runner instances), so the run-to-run difference is the instance-to-instance noise of this measurement.

Distance to rho

In this run's own matched rT770 rho (20 processes, mean cold 2,610.5 ms, step 275.68 ns), the end-to-end cold gap is **4.0999 for side A (M38) and 2.0485 for side B (D4)** (point estimates, no interval computed for the gaps in this run); S end to end cold is 0.4194 for A and **0.2095 for B** in rT770 steps `[pcc:unverified]`. These are in-run gaps on an Intel Xeon Platinum 8370C. The field-ops run read 4.096 for the same M38 code on AMD EPYC 9V74 and the field-ops-2 run 3.165 for R1 on another panel and runner: gaps do not compose across runs (the matched rho's cold mean is a mean of random-length processes and moves with panel and host), only paired deltas within one run are deltas. Wall time, one host class, factor-base construction charged to neither side, so cold S is a lower bound.

Section 5a (priced native work; a model band)

`band_model.py` (frozen) with rho counted in this run and the IC arms modelled from the archived counted run 39882571652 (per-probe and per-attempt composition, this run's measured probes and attempts, each base's exact entries): at 0, 0.1 and 1 GAE per unpriced native operation, the gap to rT is A (M38) 0.000332, 1.136 [0.8876, 1.518], 4.527 and B (D4) **9.771e-05, 0.7272 [0.5657, 0.9728], 2.897**. **D4 is cheaper than M38 at every weight** (the order holds at both ends of every bound); **against the matched rho both are unresolved**: the order flips inside the band (D4 is below rho at w = 0 and w = 0.1, and above it at w = 1; M38 is below at w = 0 and above at w = 0.1 and 1). This is a derived model, not a counted run of the field-ops producer; the counts it uses are exact per (point, seed, base) so the band is the same in the first run; the linear algebra and factor-base construction are not charged and the intervals carry no model uncertainty `[pcc:unverified]`.

Limits of this result

- Wall time on one GitHub-hosted runner (Intel Xeon Platinum 8370C), one pinned CPU, Tier 1 instance scope, no host-wide exclusivity; not replayed on another host class (the field-ops and field-ops-2 runs were on AMD EPYC 9V74).
- **Not stacked on R1.** D4 was measured against the field-ops producer on the M38 base, the row that preceded field-ops-2 (R1, `koblitz-n83-fieldops2-20261107`, run 42424680756, 1.0256 [1.0236, 1.0278] over the same field-ops M38 reference). R1 x D4 was not run, and ratios from different runs do not compose.
- Per-base identities of D4 were frozen from local runs of the same producer (reproducibility, not independent proof); the independent checks are the pinned-oracle verification of header and table and the oracle replay of every recovered scalar.
- The narrowed panel scan, the reseed and the A/A rule change are disclosed above.
- Factor-base construction is charged to neither side, as in every n = 83 row.

Files (result)

`run_github_42425640847/` (receipt, `band/`), `figures/make_result_figure.py`, `figures/result.svg`, `figures/result.png`, `figures/make_report_pdf.py`, this report as `README.pdf`.

Design in brief

Both sides run one frozen binary (`ic_online_fieldops_r770`). A = M38 closure base (K' = 16, 608 orbits), B = D4 (K' = 4, 400 orbits, `make_closure_base.py --columns 4 --norm 165`). 20 new held-out points x 3 alternating primary pairs, 2 A/A pairs and 1 planted pair per point, one rT770 per point, 60 distinct rank seeds (2501-2560); decision frozen before timing: improvement iff the 95% two-stage paired bootstrap lower bound of E2E(A) / E2E(B) exceeds 1. The full frozen design, power check, gates and disclosures (narrowed panel scan, reseed, A/A rule change) are in the experiment README.